AIMS
Airbus Material Specification

Plain weave fabric/180°C curing class
Standard modulus fibre

Material Specification

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1 Scope

This Material Specification defines the requirements of carbon fibre reinforced plain weave fabric prepreg, curing at 180°C, for aerospace application.
Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the references documents.

2 Application

Manufacture of structural components, temperature range -55°C to +95°C. Materials to be delivered according to the relevant ABS, cured ply thickness 0,205 ± 0,010mm, fibre volume fraction 53 ± 5 Vol. %.

3 Normative References

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS05-01-000 Part 1	Technical Specification - Carbon fibre reinforced epoxy prepreg - General requirements
AIMS05-01-000 Part 3	Technical Specification - Carbon fibre reinforced epoxy prepreg - Qualification and batch release testing. Fabric/180°C curing class
AIMS05-01-000 Part 100	Technical Specification - Carbon fiber reinforced epoxy prepreg - Structural material - Acceptance Testing and Storage Life Prolongation

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Specification values are minimum except where stated.

4.1.1 Material description

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4.2.2 Mechanical properties

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5 Designation

See relevant ABS.

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS05-01-000.

7 Incoming inspection

Materials qualified against this AIMS shall undego “Technical incoming inspection” as per AIMS05-01-000 Part 100 prior to use.

Table 1: Material description

Characteristic	Requirements
Material type	PW carbon fibre fabric/epoxy resin
Formulation	Fibre: standard modulus carbon fibre Resin: Epoxy, toughened, controlled flow
Form and method of production	Prepreg, cured in an autoclave at 180°C
Technical Specification	AIMS05-01-000 Part 1 and Part 3
Resin content by weight, uncured	40% nominal
Service temperature range	-55°C up to 95°C

Table 2: Worklife and storage conditions

No.	Properties	Shop condition	Symbol	Unit	Requirement
1	Storage life at -18°C	-	-	months	≥ 12
2	Storage life at 5 ± 2°C	-	-	months	-
3	Worklife [1)] (tack life)	30°C / 75% R.H.	-	days	≥ 10
		25 ± 2°C / 45 ± 10% R.H.	-	days	≥ 15

[1)] The prepreg can be exposed an additional minimum of 5 days at the above mentioned shop conditions, provided that the lay-up is fully bagged and ready for use

Table 3: Processing conditions

Cure type	Autoclave		
Cure conditions	Heating rate	°C/min	0.2 – 5.0
	Cure temperature	°C	175-190
	Cure time	min	max. 210
	Cure pressure	MPa	0.35 – 7.2
		bar	3.5 – 7.2
	Cooling rate	°C/min	0.2 – 3.5
	Vacuum (abs)	MPa	0.005 – 0.1

Table 4: Physical properties

No.	Properties	Test method	Unit	Requirements
Uncured				
1	Prepreg areal weight	EN 2557	g/m^2	325 ± 25
2	Fibre areal weight	EN 2559	g/m^2	193 ± 8
3	Resin density	ISO 1183 Method A	g/m^3	See relevant IPS
4	Fibre density	ISO 10119	g/m^3	1.78 ± 0.04
5	Volatile content	EN 2558	%	≤ 2
6	Resin content	EN 2559	%	40 ± 2
7	Physic-chemical tests	AIMS05-01-000 part 1	-	See relevant IPS
8	Resin flow	EN 2560	%	See relevant IPS
9	Tack	To be agreed between manufacturer and Airbus/Maap	-	See relevant IPS
Cured				
10	Coefficient of thermal expansion	ASTM E831	-	See relevant IPS
11	Water uptake	EN 3615, 70°C / 85% R.H.	%	≤1.5
12	Water uptake	EN 2378, 70°C / water	%	See relevant IPS
13	Extent of cure	AITM 3-0008	%	≥ 90
Note: The stated requirements are average values				

Table 5: Mechanical properties

No.	Properties	Test Method	Units	Requirements [1]																	
				Dry								Wet 70°C / 85% R.H. [3]						Water 70°C [4]			
				-55°C		RT		90°C		120°C		70°C		90°C		120°C		70°C		90°C	
				X	X min	X	X min	X	X min	X	X min	X	X min	X	X min	X	X min	X	X min	X	X min
1	Glass transition temperature, Tg onset	AITM 1.0003	°C	Xmin= 150						Xmin= 120						-					
2a	Tensile strength (warp)	EN 2561 type B	MPa	640	600	640	600	640	600	640	600	640	600	640	600	640	600	640	600	-	-
2b	Tensile modulus (warp)	EN 2561 type B	GPa	62	60	62	60	62	60	62	60	62	60	62	60	62	60	62	60	-	-
3a	Tensile strength (weft)	EN2597 type B	MPa	640	600	640	600	640	600	640	600	640	600	640	600	640	600	640	600	-	-
3b	Tensile modulus (weft)	EN2597 type B	GPa	62	60	62	60	62	60	62	60	62	60	62	60	62	60	62	60	-	-
4a	Compression strength (warp)	EN 2850 type B	MPa	775	700	775	725	675	600	625	560	600	570	550	500	425	375	500	450	-	-
4b	Compression modulus (warp)	EN 2850 type B	GPa	54	50	54	50	54	50	54	50	54	50	54	50	54	50	54	50	-	-
5	Compression strength warp quality control	EN 2850 type B	MPa	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
6a	Compression strength (weft)	EN 2850 type B	MPa	775	700	775	725	675	600	625	560	585	515	550	500	425	375	500	450	-	-
6b	Compression modulus (weft)	EN 2850 type B	GPa	54	50	54	50	54	50	54	50	54	50	54	50	54	50	54	50	-	-
7a	In plane shear strength	AITM 1.0002	MPa	105	90	115	100	90	80	80	70	93	88	75	70	60	50	80	75	-	-
7b	In plane shear modulus	AITM 1.0002	GPa	5.0	4.5	4.5	4.1	4.0	3.5	3.3	2.5	3.6	3.1	3.2	2.8	2.2	2.0	3.2	2.8	-	-
8	Compression after impact [2]	AITM 1.0010	MPa	-	-	225	200	-	-	-	-	-	-	-	-	-	-	-	-	-	
9	G_{1C}	AITM 1.0005	J/m^2	-	-	480	425	-	-	-	-	-	-	-	-	-	-	-	-	-	-
10	G_{2C}	AITM 1.0005	J/m^2	-	-	1800	1500	-	-	-	-	-	-	-	-	-	-	-	-	-	-

(Continued)

Table 5: Mechanical properties

No.	Properties		Test Method	Units	Requirements [1]																			
					Dry								Wet 70°C / 85% R.H. [3]						Water 70°C [4]			
					-55°C		RT		90°C		120°C		70°C		90°C		120°C		70°C		90°C	
					X	X min	X	X min	X	X min	X	X min	X	X min	X	X min	X	X min	X	X min	X	X min
11a	Tension strength (unnotched) [6]		AITM 1.0007 A [7] quasi-isotropic lay-up	MPa	-	-	425	380	-	-	-	-	460	410	460	410	-	-	-	-	-	-
11b	Open hole tension strength (notched) [5) 6)]		AITM 1.0007 A [7] quasi-isotropic lay-up	MPa	-	-	240	215	-	-	-	-	260	230	260	230	-	-	-	-	-	-
12a	Compression strength (unnotched) [6]		AITM 1.0008 A [7] quasi-isotropic lay-up	MPa		-	375	350	-	-	375	350	375	350	375	350	-	-	-	-	-	-
12b	Filled hole compression strength (notched) [5) 6)]		AITM 1.0008 A [7] quasi-isotropic lay-up	MPa	-	-	315	280	-	-	315	280	315	280	315	280	-	-	-	-	-	-
13a	Bearing strength	Ult.	MPa	(MPa)	-	-	575	520	-	-	-	-	460	420	420	380	360	330	-	-	-	-
		2% elong					550	505					430	400	415	375	350	320				
14	Interlaminar shear strength (warp)		EN 2563	MPa	65	60	70	65	55	50	53	48	51	47	44	40	32	29	49	45	-	-
15a	Interlaminar shear strength (reference)		EN 2563	MPa	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
15b	Interlaminar shear strength (skydrol, 70°C/1000 h)		EN 2563	MPa	-	-	70	65	55	50	-	-	-	-	-	-	-	-	-	-	-	-
15c	Interlaminar shear strength (skydrol/water, 50°C/1000 h)		EN 2563	MPa	-	-	70	65	40	35	-	-	-	-	-	-	-	-	-	-	-	-

(Continued)

Table 5: Mechanical properties (concluded)

No.	Properties	Test Method	Units	Requirements [1]																	
				Dry								Wet 70°C / 85% R.H. [3]						Water 70°C [4]			
				-55°C		RT		90°C		120°C		70°C		90°C		120°C		70°C		90°C	
				X	X min	X	X min	X	X min	X	X min	X	X min	X	X min	X	X min	X	X min	X	X min
15d	Interlaminar shear strength (fuel, RT/1000 h)	EN 2563	(MPa)	-	-	70	65	50	45	-	-	-	-	-	-	-	-	-	-	-	-
15e	Interlaminar shear strength (MEK, RT/1 h)	EN 2563	(MPa)	-	-	70	65	-	-	-	-	-	-	-	-	-	-	-	-	-	-
15f	In plane shear strength (phenolic paint stripper)	AITM 1.0001	(MPa)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
15g	In plane shear strength (non-phenolic paint stripper)	AITM 1.0001	(MPa)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

[1] The stated requirements are average values

- X means minimum average value
- X_{min} means minimum individual value

[2] BVID approximately 0,3mm

[3] According to EN 2823

[4] According to EN2378

[5] Requirements using the total width of the specimens, without discounting the hole diameter

[6] Specimen width : 38mm

[7] The lap up for these tests is $[(+0-90)_2 + 0]_s$

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 11/99		New standard
2 07/13	Chapter 7 Chapter 3 Whole document	Chapter 7 defined in line with incoming inspection (Part 100) implementation. AIMS05-01-000 Part 100 added. Changes to bring in line with standardization rules.